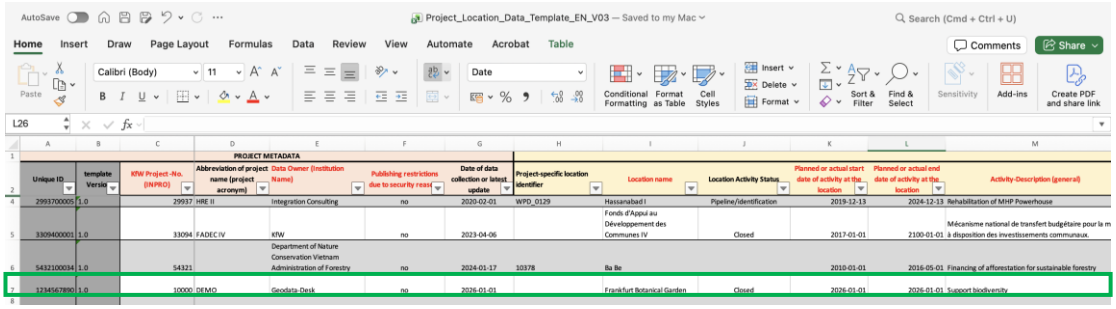
Project Locations as polygons

Reference: https://mapme-initiative.github.io/project_location_model/collection.html

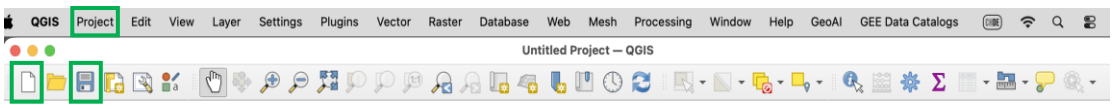
Certain Project Locations are represented as lines or polygons, rather than as points. In these cases, geometries in GeoJSON format need to be provided, in addition using the Project Location Model (Excel template) to store associated metadata.

Geometries of Project Locations can be created with Geographic Information System (GIS) software. The following steps are described using open source QGIS software.

1. Begin by collecting data using the Project Locations Model (Excel template) or PLM. For demonstration purposes, we will create a geometry for the Frankfurt Botanical Garden (Unique ID value "1234567890"), introduced here as an example.



2. Start a new project in QGIS by selecting either the New Project icon or choose **Project > New** in the menu bar. Select the Save icon or **Project > Save** to create or save the file.



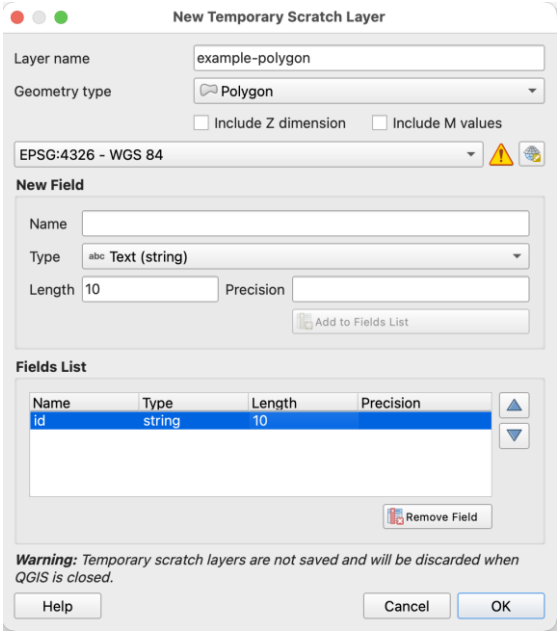
3. Go to **Layer > Create Layer > New Temporary Scratch Layer**

   - Add a Layer name (e.g. example-polygon)
   - Set Geometry type to **Polygon**
   - Set the Coordinate Reference System (CRS) to **EPSG:4326 - WGS84**, which is required for GeoJSON format

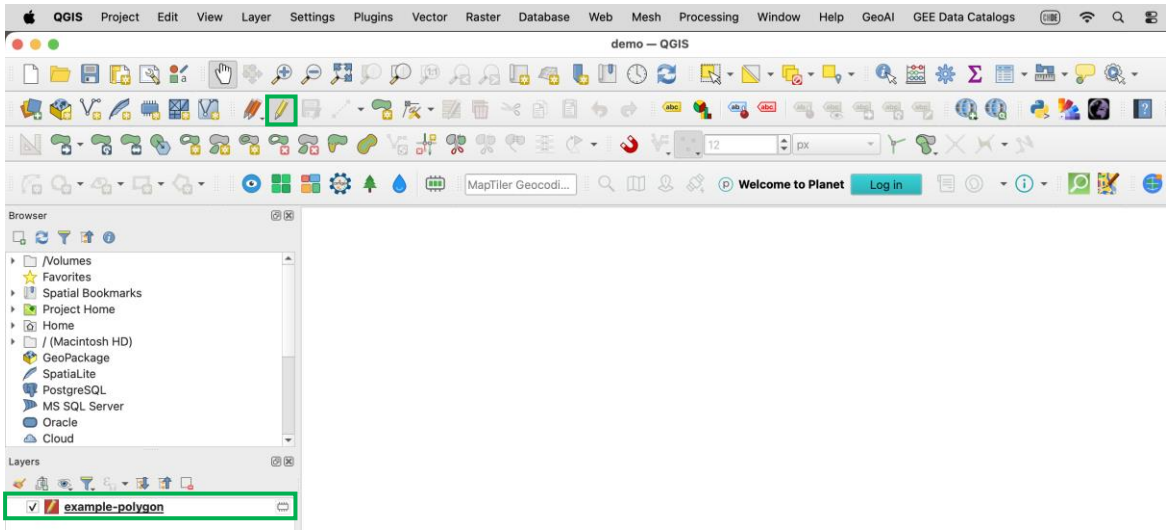
- Create at least one attribute field (e.g. “id”) and select **Add to Fields List**. This attribute field should contain the same metadata value(s) that also appear in the PLM, so that a join can be established between the two data files. While the name of the attribute fields can be different, the value(s) contained in the fields should match exactly. For example, attribute fields named “id” (in GeoJSON) and “Unique ID” (in PLM) both contain the value “1234567890”, which represents a specific Project Location.

  Note: The value(s) in both attribute fields to be joined must be exact matches:
  - same data type defined (e.g. both Text (string) or integer values)
  - same case and no extra spaces
  - ideally, values should be unique or have a 1:1 correspondence

- Once the attributes are defined, select **OK** to confirm

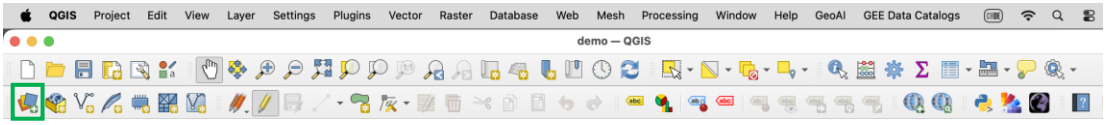


4. The new layer will appear with the editing pencil enabled in the table of contents. The **Toggle Editing** feature can also be activated and deactivated in the toolbar.

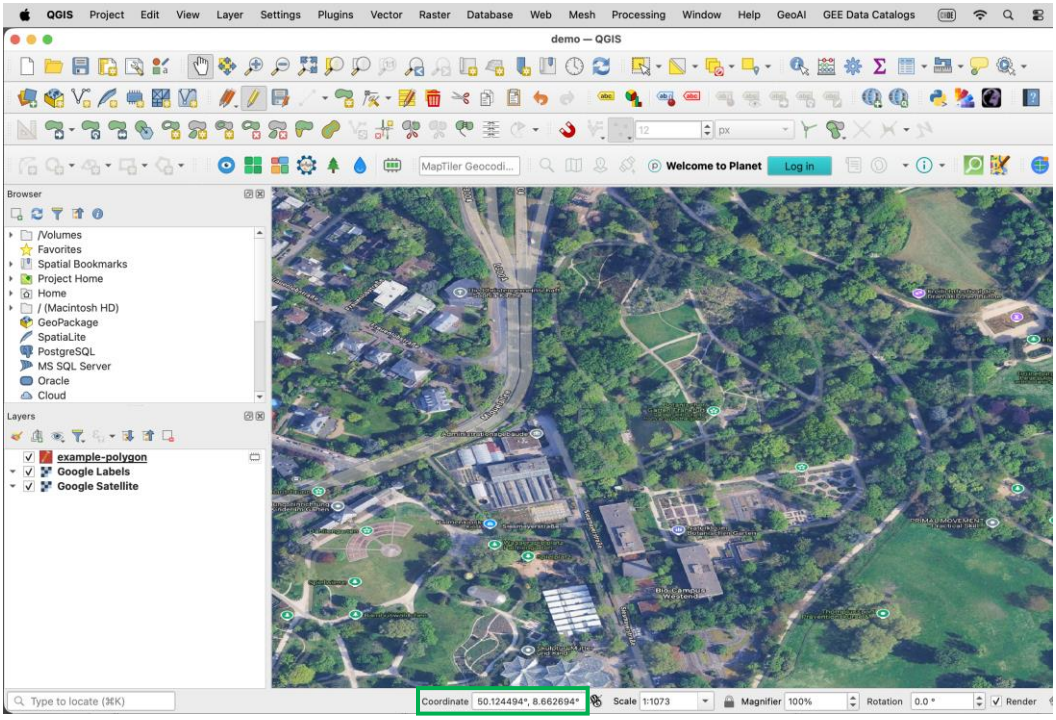


5. It generally helps to add a reference dataset to assist with digitizing or drawing the geometries.

   To add an existing data layer, select **Layer > Data Source Manager** or the **Open Data Source Manager icon**.

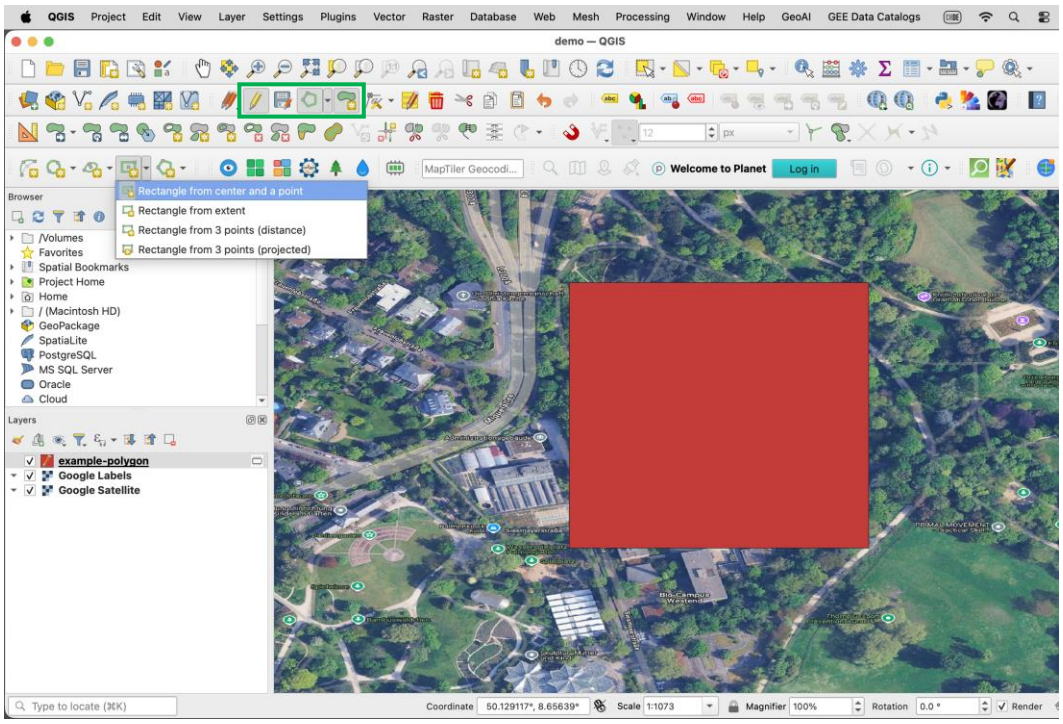


   To add e.g. a satellite basemap, select **Web > QuickMapServices** > **Google Satellite** (if you don't see this option in the **Web** menu, activate the *NextGIS QuickMapServices* plugin in **Plugins > Manage and Install Plugins**, and then in the plugin settings, click on **More Services > Get contributed pack**). Once selected, the layer will display in the table of contents as a new entry. Zoom to the area where you would like to draw the geometry or enter coordinates in decimal degrees in the **Coordinate** field at the bottom of the window for orientation. In this example, we will digitize the Frankfurt Botanical Garden located at (50.124, 8.650).



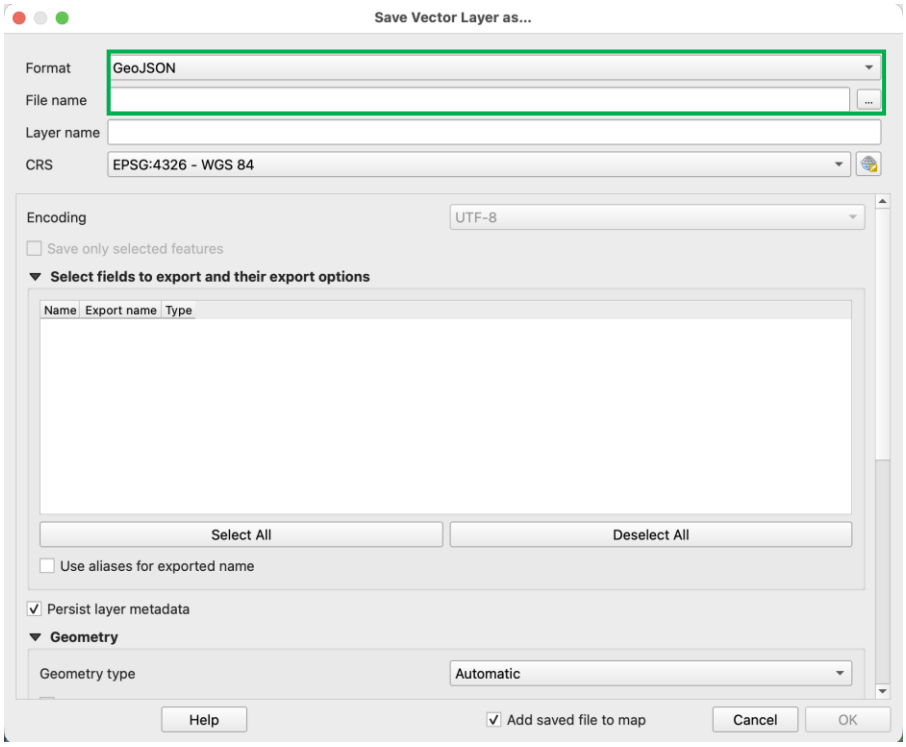
6. Select the temporary scratch layer in the table of contents. Select the **Add Polygon Feature icon** in the toolbar. This will enable a wide range of options to draw a polygon (if you don't see these options, you can enable the **Shape Digitizing Toolbar** by right-clicking the toolbar and selecting the correct Panel). Depending on the precision of the area being drawn, the polygon can be drawn as e.g. a circle or a rectangle, or as a custom multi-vertex geometry. In this example, the approximate area containing the garden is represented by a rectangle. Once you have finished drawing the polygon, right-click to close it. You will then be prompted to enter the attribute values for the fields you created earlier. After assigning the correct attributes, click **OK** to save them.

   Use **Layer > Save Layer Edits** frequently while digitizing.

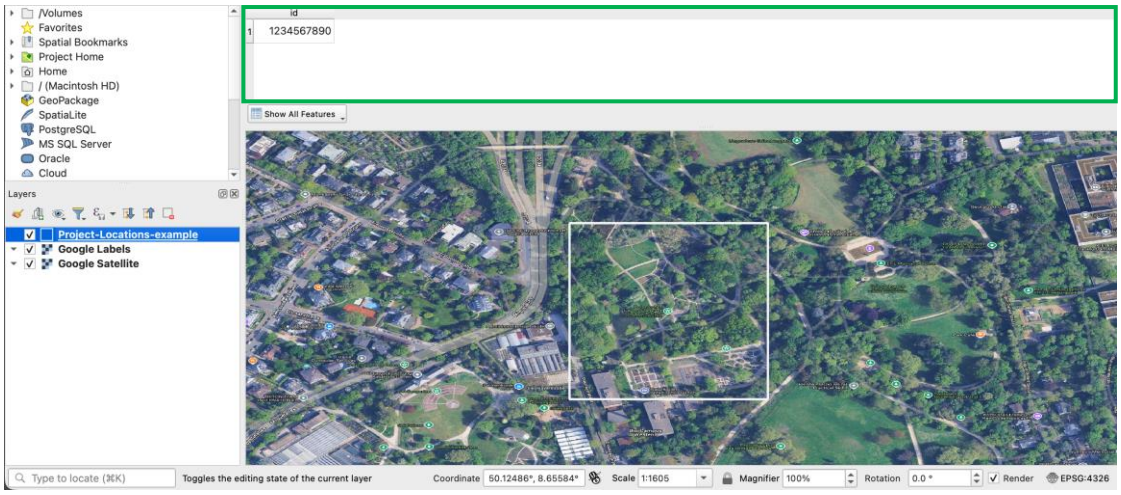


7. Repeat Step 6 to create a new geometry for each Project Location.

8. Once you have finished adding geometries, click on the pencil icon in the toolbar to stop editing the layer. Right-click on the temporary layer > **Export > Save Features As** to save it permanently. Set format to **GeoJSON**, choose output path, then click **OK**. The new permanent data layer with the Project Locations geometry will appear in the table of contents afterwards. You may remove the temporary layer at this point.

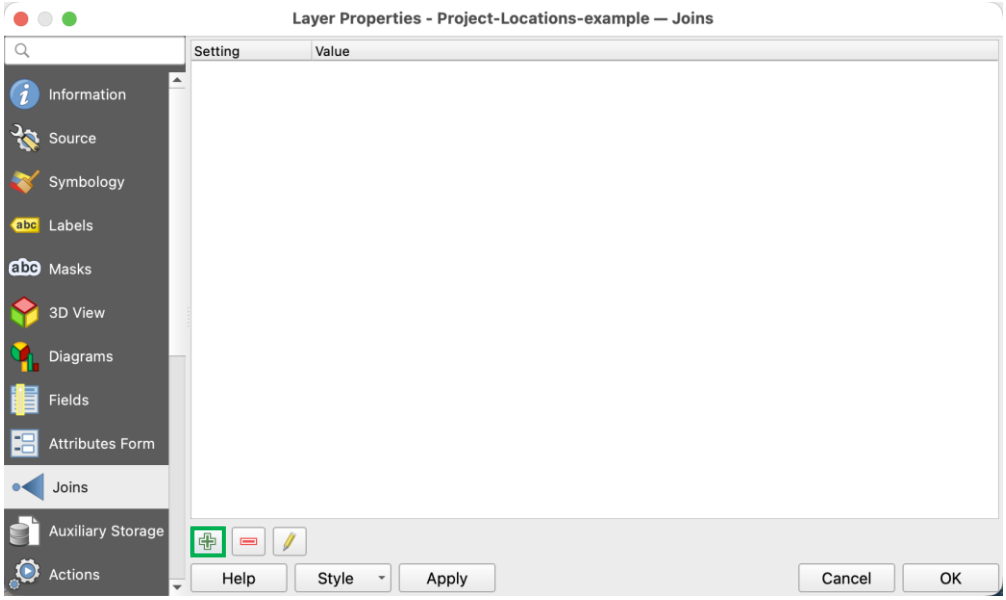


9. Right click on the GeoJSON data layer in the table of contents > **Open Attribute Table**.

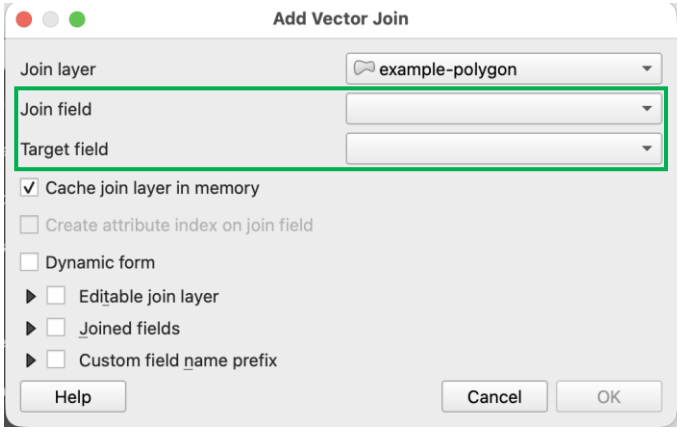
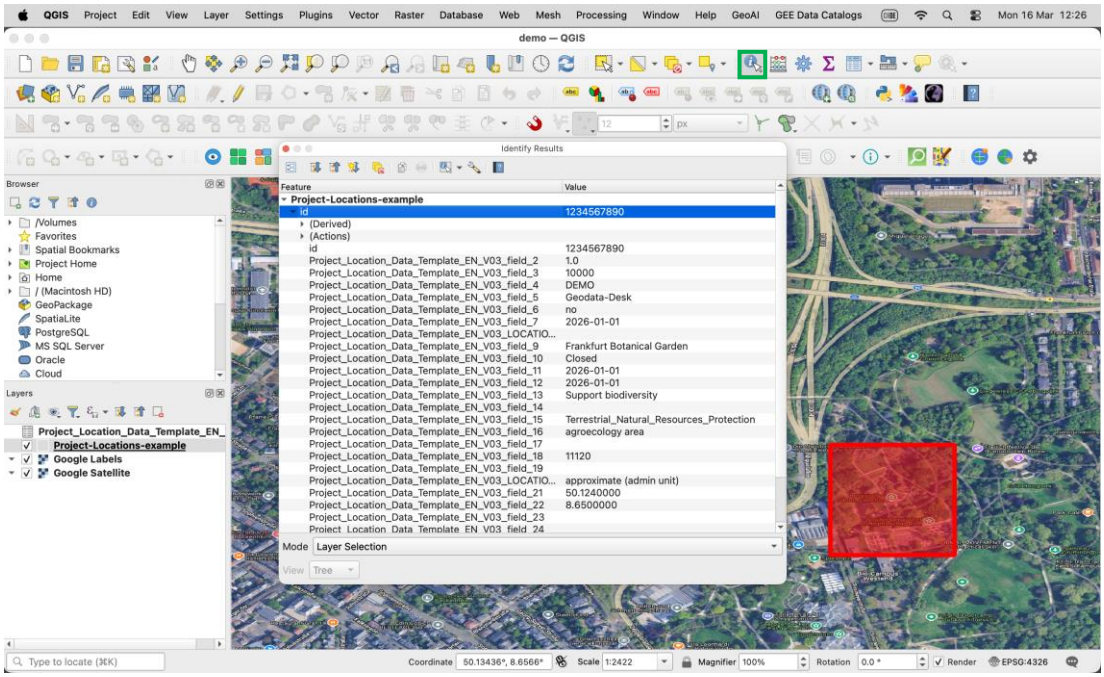


10. Add the PLM (Excel table) by dragging the file from Windows File Manager or Finder into the QGIS table of contents area.

11. Right-click on the GeoJSON layer in the table of contents and select **Properties** to open the **Joins** tab and click on the **+** button to add a join.



12. Select the Project Locations Model (Excel table) as the join layer. Choose the **join field** (from Excel table) and the **target field** (from GeoJSON). Click **OK** to confirm.



13. To verify the newly created data layer, open the **Attribute Table** of the GeoJSON data layer and check the **Layer Properties > Information** tab. The metadata from the PLM should appear. This can also be verified by clicking on the polygon that was created, after selecting the **Identify Features** icon.